

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2013-14

Paper No. BPT-201: Thermal Physics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. Answer any **four** of the following: [4 × 3.5 = 14]

- (a) Find the probability of an oxygen molecule having its velocity in the range 99.5 to 100.5 m/s at 200 K. Given: molecular weight of oxygen = 32 and Avogadro number $N_A = 6.023 \times 10^{23}/\text{mol}$.
- (b) Find the change in entropy of the universe in converting 1 g of ice at 0°C into steam at 100°C. (Latent heat of fusion of ice = 80 cal/g, latent heat of vaporization = 540 cal/g).
- (c) Show that Gibbs function remains constant for an isobaric process carried out in such a way that the final temperature remains equal to the initial temperature at the conclusion of the thermodynamic process.
- (d) Discuss whether blackbody radiation inside an enclosure remains black even after adiabatic expansion.
- (e) The efficiency of a Carnot engine working in a reversible cycle between a source and a sink (temperature of sink = 127°C) is 1/6. If the temperatures of source and sink are reduced by 100°C, find the efficiency of the Carnot engine.

Question 2.

- (a) How are absolute zero and unit degree of temperature defined on the thermodynamic absolute scale of temperature? [5]
- (b) Discuss how Andrews' experiments lead to the existence of critical constants for carbon dioxide gas undergoing liquefaction. [9]

Question 3.

- (a) Show that the correction for volume due to the finite size of molecules for a van der Waals gas is four times the actual volume of all the molecules. [5]
- (b) Show that there is an increase in the entropy of the universe for an irreversible process. [4]
- (c) What are the Clausius and Kelvin-Planck statements of the second law of thermodynamics? Show that the Kelvin-Planck and Clausius statements of the second law of thermodynamics are equivalent. [5]

Question 4.

- (a) Show that for an ideal gas, internal energy is a function of temperature only and is independent of volume, while for a van der Waals gas it depends on volume also. [5]

OR

- (a) Find an expression for the fall in temperature resulting from the adiabatic stretching of a liquid film, if there is a small increase in the surface area of the film and the volume of the film remains nearly constant. [4]

- (b) Show that:

$$C_p - C_v = T \left(\frac{\partial p}{\partial T} \right)_v \left(\frac{\partial V}{\partial T} \right)_p$$

and hence deduce that for a van der Waals gas:

$$C_p - C_v \approx R \left[1 + \frac{2a}{VRT} \right]$$

where the symbols have their usual meanings. [6]

- (c) Derive the Clausius-Clapeyron equation using the appropriate Maxwell's thermodynamic relation. With the help of this relation, discuss the variation of boiling and melting points with pressure. [4]

Question 5.

- (a) Explain the phenomenon of adiabatic demagnetization of a paramagnetic salt. How will you employ this phenomenon to produce very low temperatures? Is absolute zero attainable? [8]

- (b) Show that the Joule-Thomson expansion is an isenthalpic process and derive the expression for the Joule-Thomson coefficient for a van der Waals gas:

$$\mu_{JT} = \left(\frac{\partial T}{\partial p} \right)_H = \frac{1}{C_p} \left[\frac{2a}{RT} - b \right]$$

where the symbols have their usual meanings. [6]

OR

- (a) Show that the perfect gas scale and the absolute thermodynamic scale are identical. [6]

- (b) Derive the first and second $T dS$ equations using Maxwell's thermodynamic relations. [4]

- (c) What is superfluidity with reference to liquid Helium? Discuss the phenomenon of second sound and the velocity of second sound. [4]

Question 6.

- (a) Obtain the Rayleigh-Jeans formula for the distribution of intensity in the emission spectrum of a black body. Comment on its success and failure. [10]
- (b) Define Boyle temperature. What is the relation between Boyle temperature, critical temperature, and temperature of inversion? [4]